

PIR - SAP POSDTA application was inaccessible on 28/02

M&S Digital&Tech *Post Incident Review*



Incident/ Problem Reference: 90634845/ 71723

Incident Date: 28/02/2025

PIR Date: 06/03/2025

Author: Saloni Chowdhury

Technical Owner: Vidhyasagar Balakrishnan

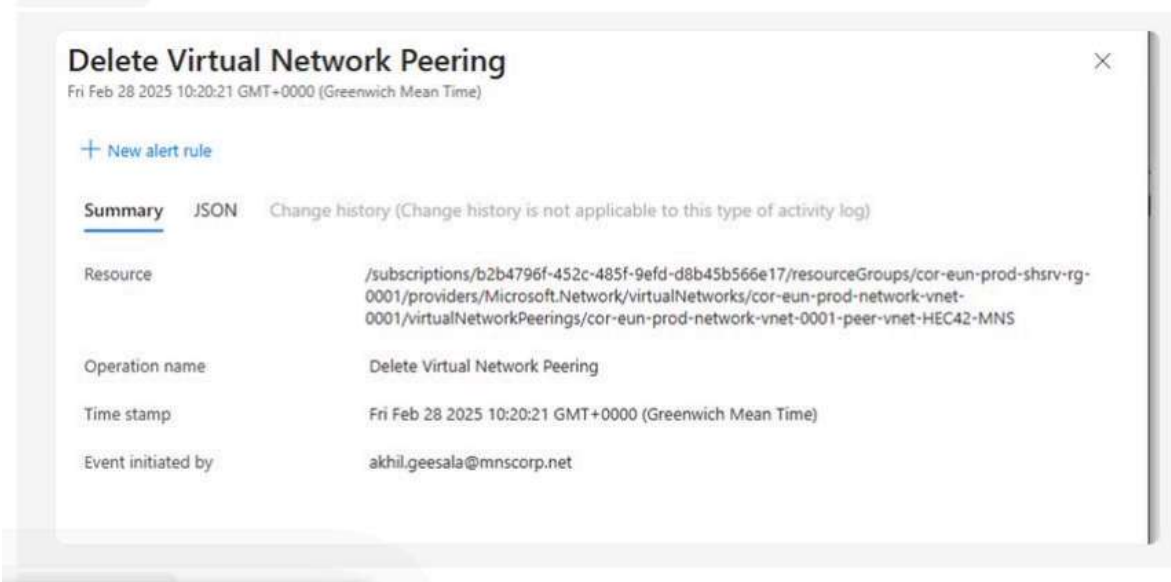
PIR Reviewer: Anna Scaria/ Gopi Kishan

1. Incident Summary :

On 28/02, SAP POSDTA was inaccessible between 10:30 and 13:00 impacting POS sales feeds, GMOR feeds to the downstream systems, however, with no business impact. The root cause was attributed to an inadvertent deletion of Vnet peer within Azure network as part of a security housekeeping activity. The Vnet peer was re-instated to restore the services by 13:00..

2. Incident Chronology :

11:31 It was identified that the Vnet peering was deleted (we are yet to get when and how). Event identified in the log



- 11:35 Vendor SAP engaged to authorize the newly created Vnet to restore service - work in progress.
- 11:40 Vendor SAP has joined the MIM call to assist to authorize the newly created Vnet to restore services
- 11:40: As there is no business impact until 14:00, text and comms are being held.
- 11:58 TSM to request SAP to provide the authorization for Fastin to establish the peering.
- 12:02 The issue was identified when the Control-M jobs failed as one of the Control agents was down.
- 12:03 SAP to provide authorization to M&S network to initiate the Vnet peering. SAP to get the authorization through their internal teams
- 12:08 Considering Fastin is not in SAP system's active directory it is taking time to get authorization, discussion is underway to identify the feasibility of having Fastin as a guest user to get the access.
- 12:11 SAP Active directory team is being engaged to add Fastin and then to provide authorization.
- 12:15 SAP team is working on finding right type to invite that can be sent to Fastin to get the access in to SAP
- 12:24 SAP has advised that Fastin should receive the access request invite in 2 mins
- 12:26 Access has been enabled for Fastin and he is yet to receive the link from SAP to get through their system.
- 12:35 Fastin has got the access to SAP system and logged in to the SAP azure portal.
- 12:39 The issue was caused due to a manual error where the peering was deleted
- 12:49 We are encountering authorization related error while adding the peering in SAP, further investigation continues.
- 12:58 The peering has been reinitiated after it was created in both SAP and M&S end to restore the services.
- 13:00 SAP POSDTA both the prod and non-prod is accessible,
- 13:08 SAP ECC, Sales team are validating the held jobs to
- 13:09 As part of deleting the stale Vnet peering, the team has deleted the SAP POSDTA peering incorrectly due to a oversight.

3. Incident Resolution

👉 *The actions taken to restore services and mitigate impact.*

The impacted VNER peer was reinstated to restore POSDTA application availability.

4. Root Cause

The root cause was attributed to an inadvertent deletion of VNet peering under SAP subscription due to lack of impact assessment and process gap during a security cleanup activity of stale VNETs within the Azure network layer.

Note: A VNet (Virtual Network) in Azure is a private, isolated network that allows Azure resources to securely communicate with each other, the internet, and on-premises networks.

Context:

As part of security measures, the cybersecurity team flagged stale VNet subscriptions for cleanup. During this process, the Cloud Frameworks team identified SAP VNet peers.

As per the standard process, when a stale subscription is deleted, its associated VNet peers should also be removed from Azure cloud Network layer - this was not followed resulting in the inadvertent deletion.

While performing the cleanup, although the cloud engineer had access to both source and destination VNets, the SAP subscription—being a SaaS solution—the team lacked visibility into its functionality and mistakenly assumed it was stale without performing proper impact assessment, leading to its deletion.

To prevent such incidents, we are reviewing the stale subscription decommissioning process to ensure VNets are not deleted ad hoc. Going forward:

- **Impact Assessment:** Relevant owners must be identified for each subscription to evaluate potential impact before deletion.
- **Third-Party Subscriptions:** A documentation exercise will be conducted for all third-party subscriptions within Azure to assess their status before any deletion.
- **Preventive Controls:** We have identified a locking mechanism to avoid inadvertent deletions.
- **Firefighter Access & Privileged Identity Management (PIM):** Efforts are underway to secure Azure PIM (Privileged Identity Management) privileges for SAP accounts to expedite restoration in case of accidental deletions. The SAP vendor had revoked firefighter access for the cloud and network teams, and discussions are ongoing to reinstate it.
- **Change Management Compliance:** The deletion of VNet peers in production was performed without a change record, deviating from the change management process. The cloud team has been advised to strictly follow the change management protocol for all future activities

5. Findings .

5.1. Observability –

The issue was identified through a control and job failure within POSDTA when a sales job failed to start due to application unavailability. The SAP product team attempted to log into the SAP POSDTA application but was unable to, which led to the incident being discovered.

Regarding alerting, the necessary alerts are in place. However, it is not feasible to configure alerts for inadvertent VNet subscription deletions, even if the underlying application linked to the VNet peer is live.

5.2. Technical Challenges -

During the initial cleanup activity, the cloud team lacked visibility into the functionality and status of the SAP VNet peers, as SAP operates as a SaaS solution. This limitation prevented them from verifying whether the VNet was actively in use.

During the recovery phase, the absence of required access further delayed restoration, as the team had to wait for the SAP vendor to authorize the reinstatement of the VNet peer.

To prevent such delays in the future, we have agreed on actions to work with the SAP vendor to establish firefighter privileged access for a designated network engineer. This will help expedite recovery efforts and minimize downtime in case of similar incidents.

5.3. MIM involvement & impact assessment -

The criticality of the application's unavailability was identified promptly, and the MIM process was engaged in a timely manner. While there was no immediate business impact, as the application supports the sales flow into multiple downstream applications (GMOR, Range Planner, Sales Stock & Intake) it could have potentially affected overnight critical sales reporting.

Assessing the business impact took time, which highlighted the need for better visibility during such incidents. To address this, we have agreed with the relevant service leads to predefine impact statements for SAP POSDTA unavailability. This will help ensure a clearer and quicker understanding of the impact when similar issues arise.

5.4. Incident handling/ stakeholder management –

During incident recovery, the M&S product teams provided prompt assistance, and the SAP vendor joined the call in a timely manner. However, obtaining the necessary access for our network team to reinstate the VNet peering caused significant delays.

To mitigate this in the future, we are working to secure firefighter access, which will streamline the access procurement process and reduce recovery time in similar scenarios.

5.5. Business Communication –

There were no issues with business communications or the MIM process. All business comms were sent successfully, and the incident loop was continuously updated with real time recovery progress.

While the process functioned well, we have agreed to enhance our impact assessment and communication strategy to ensure clearer and more concise impact statements in business comms moving forward

5.6. Supplier Engagement –

The SAP vendor joined the MIM call promptly. However, due to the nature of the issue, reinstating the VNet peer within the M&S network required

additional authorization from SAP. The process was delayed as it took significant time for SAP to create the necessary AD account for the M&S network colleague, impacting recovery time.

As part of our action plan to secure firefighter access, we aim to eliminate such delays and streamline the recovery process in future incidents.

Mitigation Actions:

	Action	Owner	Due Date	Remarks	Action Category
1	Conduct a detailed review of the Azure subscription decommissioning process to prevent ad-hoc deletion of stale VNet peers. Identify owners for all subscriptions, particularly third-party ones, to enable thorough impact assessments before any deletions.	Balakrishnan, Vidhyasagar		As per the existing process, VNet peers should be deleted when a subscription is decommissioned. However, in this case, the M&S Cloud team lacked access to assess the impact due to the SaaS nature of the subscription. Going forward, we will streamline owner consultations during cleanup activities to ensure proper impact assessment and mitigate the risk of inadvertent deletions.	
2	Document all third-party subscriptions where access is unavailable to verify the status of underlying VNet peers, preventing inadvertent deletions within the Azure network layer.	Balakrishnan, Vidhyasagar Xavier, Fastin		The cloud team will identify all third-party SaaS subscriptions and their respective owners to ensure a thorough impact assessment before proceeding with any security cleanup activities.	
3	Implement a locking mechanism within the Azure network layer to prevent inadvertent deletion of active VNet peers or subscriptions.	Balakrishnan, Vidhyasagar		The cloud team is in the process of configuring this mechanism to ensure that active VNets and subscriptions cannot be deleted in error.	
4	Collaborate with the SAP vendor to procure Firefighter Azure PIM privileges by creating an AD account for a designated M&S account (ideally a service account - Y account). This will enable faster recovery in reinstating VNet peers in case of accidental deletion.	Kishan, Gopi Balakrishnan, Vidhyasagar		Discussions are underway with SAP to secure these privileges, ensuring quicker resolution and minimizing downtime during recovery.	
5	Review the overall VNet peering configuration to ensure it is managed as code within source control.			The cloud team has confirmed that VNet peering for internal M&S subscriptions are already configured as code. However, for SAP's subscription, authorization constraints exist. The team will explore options to extend this approach to third-party subscriptions.	

6	Review the security housekeeping process for deleting VNet peers or performing any activities on production subscriptions within Azure network layer.	Balakrishnan, Vidhyasagar		The VNet peer was deleted in production without a change record. The team has been advised to strictly follow the change management process for all production activities to ensure proper tracking and compliance - Completed	
---	---	--------------------------------------	--	---	--

<Mandatory participants: Put it in the meeting invite. >

Technical Details and Useful Documents: